

GradeMaster

Predictive Career Intelligence System

Software Development Cell Presentation

The Innovation Hub

Empowering Student Developers for Industry Success

Software Development Cell (SDC)

The Mission

Our College's SDC serves as a bridge between academic learning and industry standards. We foster a culture of technical excellence, focusing on building high-impact software solutions that solve real-world problems.

Incubation Ecosystem

We provide a collaborative platform where students work on full-cycle projects. Through mentorship and peer reviews, we ensure our developers are equipped with modern tech stacks and management skills.



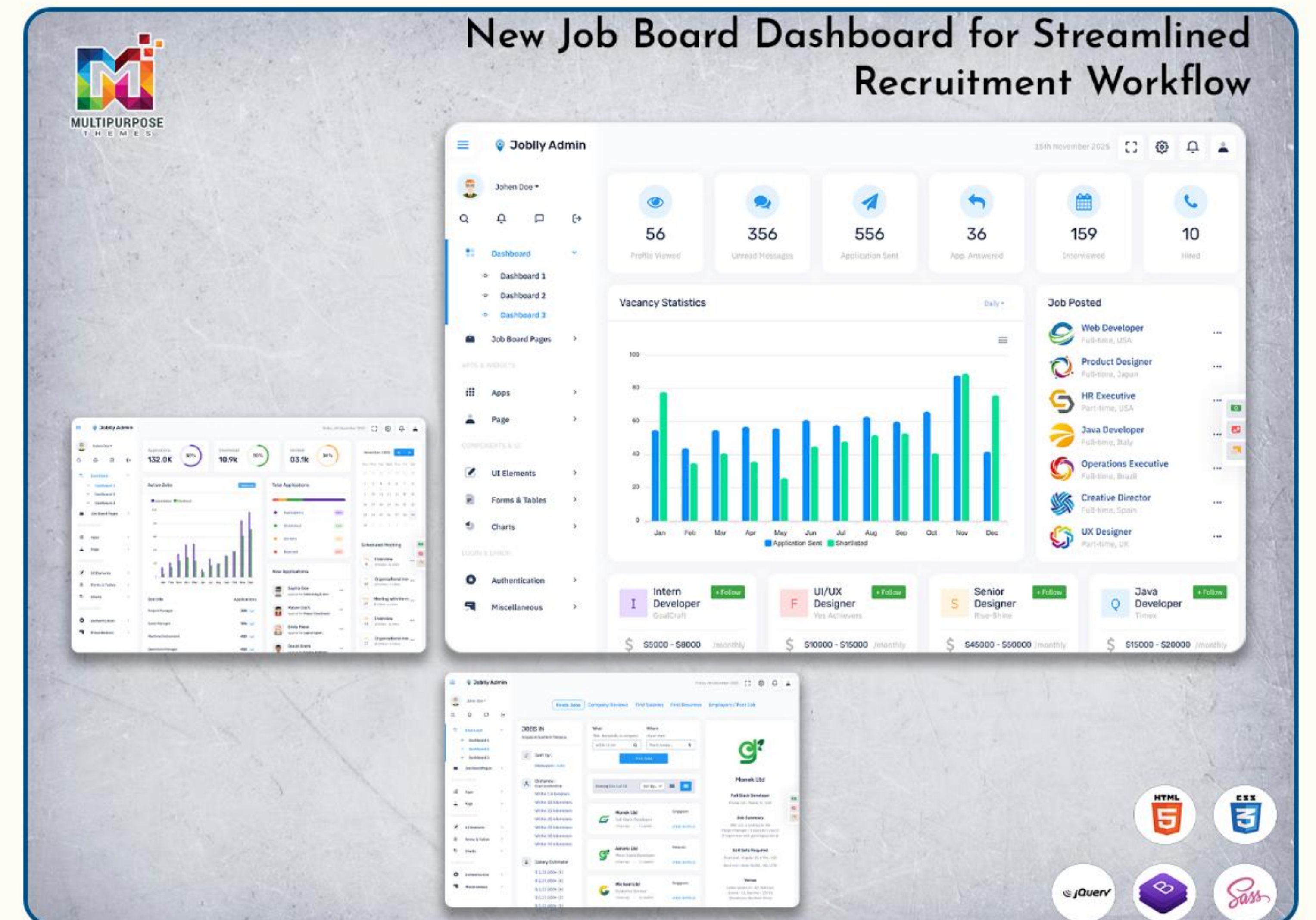
Introduction to GradeMaster



What is GradeMaster?

GradeMaster is an advanced **Predictive Career Intelligence System**. It moves beyond passive job tracking to provide candidates with a data-driven "Selection Probability" score.

By analyzing the gap between a candidate's profile and organizational requirements, it calculates proximity and guides the user toward peak selection probability.



System Architecture: Core Modules



Readiness Hub

Real-time dashboard visualizing profile strength and selection readiness.



Proximity Engine

Dual-input workspace for JD-Resume semantic alignment and heatmaps.



Interview Prep

AI Simulator utilizing the STAR framework for behavioral necessity.



Nexus Scheduler

Strategic placement roadmap and preparation block manager.



Placement Hub

Matched organizational cards with probability thresholds.

Detailed Module Descriptions – Part 1

Readiness Hub (Dashboard)

A centralized command center that displays a user's selection probability at a glance. It features the **Readiness Gauge**, which breaks down scores into Technical, Soft Skills, and Experience categories.

Selection Proximity Analyzer

The "Brain" of the project. It provides a split-pane interface where users paste a JD. The system generates a **Keyword Heatmap** and optimization tips to align the resume with organizational language.

Detailed Module Descriptions – Part 2

AI Interview Simulator

Focuses on the **Necessity phase**. It generates role-specific questions and provides feedback on clarity, structure, and technical relevance using the STAR method.

Nexus Scheduler & Hub

Scheduler: Manages the preparation timeline.

Hub: A marketplace of organizations where users meet the "High Probability" threshold before applying.

Project Milestones: Tasks Completed

Milestone 1

Landing Page: Fully responsive home page built with modern SaaS aesthetic and Electric Blue palette.

Milestone 2

Authentication: Secure Login system integrated with Postgres backend and session management.

Milestone 3

3-Step Onboarding: Multi-page registration capturing identity, tech DNA, and initial resume upload.

Integrated Tech Stack

Software / Tech	Role in Development
Antigravity	High-performance cloud-native compute and deployment environment.
PostgreSQL	Relational database for storing user profiles, JDs, and metrics.
Express.js / Node.js	Backend logic, scoring algorithms, and API orchestration.
React / Figma Make	Frontend UI implementation and high-fidelity design prototyping.

Database Design: ER Schema

Table	Primary Columns
Users	id, name, email, profile_json (DNA)
Jobs	id, company_name, jd_text, threshold
Analysis	user_id, job_id, proximity_score, heatmap
Sessions	id, user_id, star_response, feedback

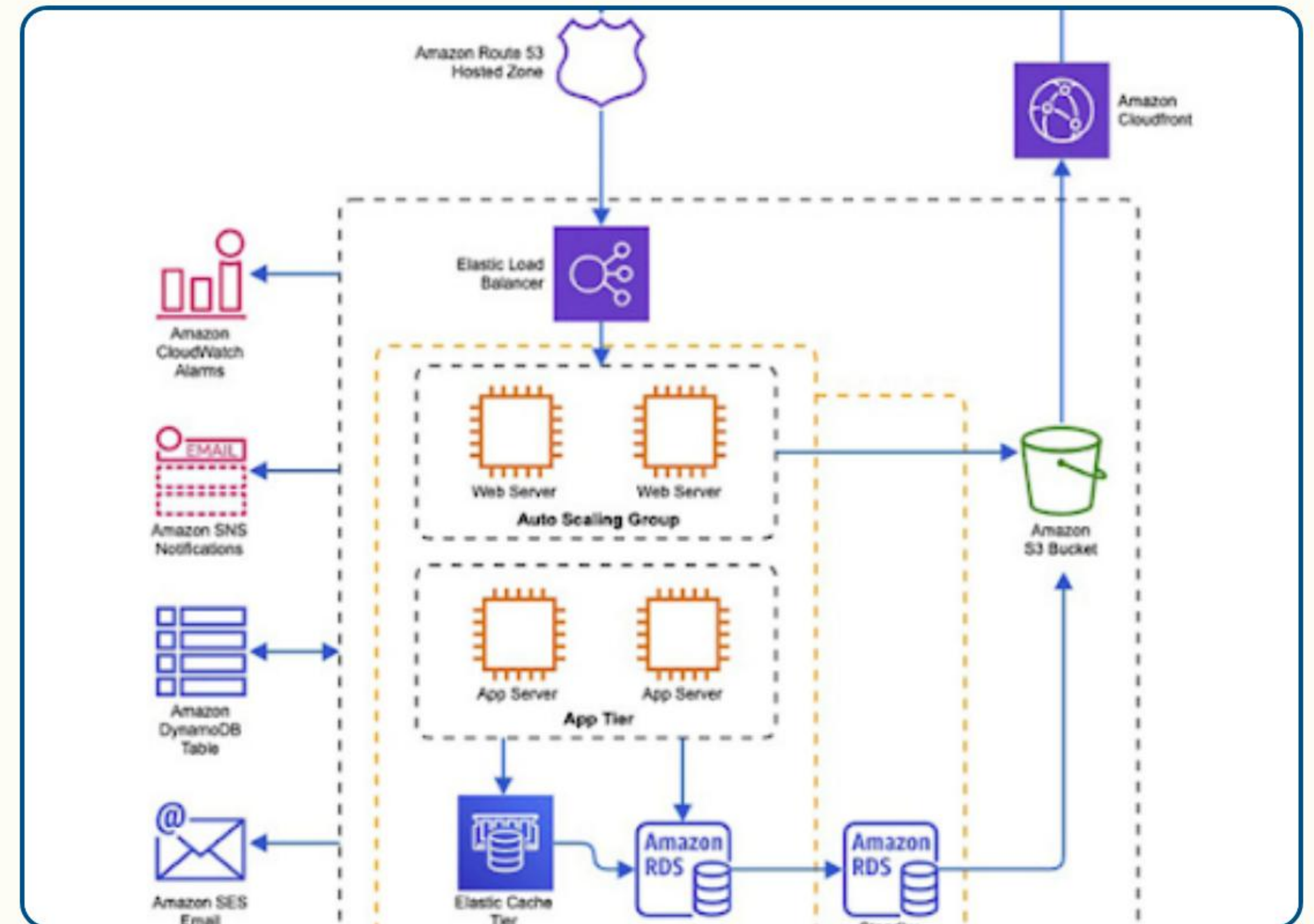


Image Sources



<https://greatwall.ma/storage/2026/02/Undergraduate-students-collaborating-in-a-modern-engineering-lab-as-part-of-the-diverse-bachelor-programs-at-HIT.webp>

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